NUMERICAL ANSWERS – CHIEF (Exam only, EXCEPT RESITS, published to students)

Question	Numerical Answer
Q1 (a)	According to lecture notes.
Q1 (b)(i)	31.52 deg (or 0.55 rad); 96%
Q1 (b)(ii)	143.25 deg (or 216.87 deg)
Q1 (b)(iii)	48.62 kWs (you only need to calculate the average power and energy)
Q2 (a)	According to lecture notes.
Q2 (b)	115.4 A (starting current); 34.8 A (full load current)
Q2 (c)	934.5 RPM (or 97.86 rad); 202.5 RPM (or 21.2 rad)
Q2 (d)	372.05 NM; 321.2 NM (comparison)
Q3 (a) (i)	$C = 0.0009\angle 90.076$
Q3 (a) (ii)	$\lvert V_{S,l-l}\rvert = 223\ kV\ ,\ \ \%reg = 8.29\%$
Q3 (a) (iii)	$\lvert V_{r,l-l}\rvert = 219.8\ kV$
Q4 (a)	$\Delta P_{g1} = 12\ MW, \Delta P_{g2} = \Delta P_{g3} = 4\ MW, \Delta_f = -0.06\ Hz$
Q4 (c)	$b = -2.3529$, $\lvert V_2\rvert = 0.915\ pu$

ASSESSMENT CHECKLIST 2

MARKS FINALISATION – CHIEF AND PROOF

We confirm that:

- The marking and checking of the above assessment have been completed in accordance with the Manual of Assessment Procedures.
- The marks conform to the advertised marks matrix, Senate Marking Scale and appropriate UG/MScT Level descriptors.
- Marks have been entered onto the spreadsheet by the Chief, moderated by the Proof and checked by the Checker.

Chief: _____Dr Mahmoud Shahbazi_____ **Date:** ____08/06/2020_____

Proof: __Dr Behzad Kazemtabrizi_____ **Date:** __08/06/20_____

Assessment Average: 70.61%

Are marks within the expected bands for this level? Yes X No □
(NOT required for resit examinations and examinations with less than five submissions)

Chief's Report (Exams and Coursework, published to students and external examiners – NOT required for resit examinations)

- Brief discussion (if necessary) as to why marks are outside expected range.
- Cohort feedback: What went well? What were the common errors or misunderstandings?

The average is at the top of the expected band, which is due to the 48 hour open book exam format.

Chair BoE: ___Claudio Balocco_____ **Date**____12/6/2020